FLA — Unit 2: Regular Expressions and Regular Languages

Regular expressions over Σ are built from ∅, ε and single symbols using union (+), concatenation and Kleene star (*). For example (0+1)*01 describes binary strings ending in 01.

Kleene's theorem

A language is regular iff some regular expression describes it iff some finite automaton accepts it. Thompson's construction turns a regex into an ε-NFA; state elimination or Arden's theorem (R = Q + RP has solution R = QP*) turns an automaton back into a regex.

Closure properties

• Regular languages are closed under union, intersection, complement, concatenation and star.
• They are also closed under reversal, homomorphism and inverse homomorphism.
Closure properties are often combined with the pumping lemma to prove that a language is not regular.